

TASMIHA HASSAN

Software Engineer

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Eligible for UK Youth Mobility Scheme and Global Talent dependant visas — no employer sponsorship required

PROFESSIONAL SUMMARY

Software engineer with 5 years building and operating large-scale cloud security products at Microsoft. Full-stack across React/TypeScript, C#/.NET services, and Scala/Spark pipelines on Azure, owning features from design specification through deployment, monitoring, and adoption metrics. Co-inventor on a pending AI agent risk patent. Relocating to the United Kingdom and available to interview at UK hours.

EXPERIENCE

Microsoft Vancouver, Canada
Microsoft Purview Insider Risk Management — enterprise data security and risk detection. Jun 2021 – Present
Software Engineer II Aug 2025 – Present

- Owned the scoring logic, validation, and hardening of an agent session scoring pipeline in a Scala/Spark distributed processing service, building on a colleague's initial skeleton to implement record-type routing, configurable insight thresholds, and time-gated state merging with expiry management
- Automated provisioning of Event Hub namespaces, Key Vault secrets, access policies, and managed identity role assignments across 20+ Azure regions in PowerShell, cutting a multi-week manual deployment to three days and producing a reusable tool for subsequent rollouts
- Authored the technical design specification and drove cross-team approval for a new risk-signal integration with Microsoft Entra, defining the event schema and implementing Graph API sync for ingesting agent detections
- Built a durable Azure Functions orchestrator producing daily adoption metrics across all tenants, feeding a Kusto dashboard reviewed weekly to locate customer drop-off in the onboarding funnel

Software Engineer Jun 2021 – Jul 2025

- Owned the Priority Content Only Scoring feature end-to-end, implementing full-stack changes from the React/TypeScript interface through C#/.NET backend scoring across 4 independent services, reducing noisy alerts for enterprise customers by 15%
- Migrated alerts, insights, and role-based access control into the Microsoft Defender portal and owned the release process on the team's Azure DevOps CI/CD pipeline, achieving 36% product adoption and 30x exposure to new users
- Migrated High Privilege Access authentication to long-running token flows across API, interface, and background processing layers, enabling zero-trust compliance and reducing credential exposure through to government cloud
- Launched browser forensic evidence recording by analysing cross-team ingestion pipelines and leveraging upstream user signals to trigger targeted URL-based application capture, expanding investigative coverage for enterprise organisations

McMaster University Hamilton, Canada
Program Support Assistant, Faculty of Engineering Jan 2021 – Apr 2021

- Led development of an accessibility-focused mobile application in React Native and Firebase, improving daily task learning outcomes for users with cognitive disabilities based on supervised user testing; presented to 100+ industry professionals and placed third globally in the Digital Inclusion competition

Embross Toronto, Canada
Software Engineer Intern May 2019 – Aug 2020

- Designed and implemented full-stack tooling for airline check-in systems in React, Node.js, Express, and SQL, reducing engineering configuration time by 90% and resolving 55+ defects across 2 airline workflows
- Gathered requirements from airline stakeholders on site, then led full-stack implementation of a frequent flyer loyalty programme enabling kiosk check-in benefits for approximately 3 million passengers

TECHNICAL SKILLS

Languages	C#, Java, JavaScript, TypeScript, Scala, Python, C, SQL, KQL
Frameworks	.NET, React, React Native, Node.js, Express, Redux, Spark, Spring MVC
Cloud	Azure (Event Hubs, Functions, Key Vault, App Service), Cosmos DB, Kusto/ADX
Practices	REST APIs, CI/CD (Azure DevOps), Git, PowerShell, automated testing, observability, technical design specifications, GDPR and EU Data Boundary compliance, accessibility (WCAG)

PATENT

Artificial Intelligence Agent Risk Assessment and Mitigation (pending) — co-inventor of a system for detecting risky AI agent behaviour in enterprise environments, enabling real-time risk assessment and policy enforcement

EDUCATION

McMaster University Hamilton, Canada
Bachelor of Engineering, Software Engineering Sep 2016 – Apr 2021
Relevant Coursework: Large Systems Design, Data Structures and Algorithms, Software Testing, Databases, Machine Learning